

TECHNICAL PROPOSAL & SYSTEM SPECS

AI-Powered Industrial Safety Intelligence for Zero-Harm Operations

A MERN-stack multi-agent safety orchestration platform designed to correlate SCADA telemetry with active work permits and prevent compound ignition and asphyxiation accidents in heavy industrial plants.

PREPARED FOR

ET AI Hackathon 2.0 (Phase 2 Submission)

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REPOSITORY URL

github.com/surjeetkumar8006/ET-AI-Hackathon-2.0

SUBMISSION DATE

June 24, 2026

AI-Powered Industrial Safety Intelligence for Zero-Harm Operations

- **Theme:** Industrial Intelligence / Worker Safety / Geospatial Safety Analytics
 - **Problem Statement:** PS 1 - AI-Powered Industrial Safety Intelligence for Zero-Harm Operations
 - **Target Audience:** Safety Officers, Plant Operations Managers, Chief Inspectors of Factories, and Industrial Executives
 - **GitHub Repository:** <https://github.com/surjeetkumar8006/ET-AI-Hackathon-2.0>
 - **Author/Team:** Surjeet Kumar (Team Lead)
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1. Executive Summary

SentinelAI is a real-time safety intelligence overlay designed to eliminate fatal workplace accidents in heavy industrial sectors (e.g., steel plants, chemical refineries, storage terminals). Traditional plants rely on SCADA systems, work permit registries, and CCTV cameras operating in isolated functional silos. As demonstrated by the Visakhapatnam Steel Plant disaster in January 2025, warning signals of gas leaks can exist inside SCADA systems, but they go unacted upon because they are not dynamically fused with active human maintenance records.

SentinelAI bridges this gap by acting as an intelligent coordination shell. It ingests:

1. Live SCADA telemetry (Methane LEL%, Oxygen O2%).
2. Digital Work Permit (PTW) registries (Hot Work, Confined Space Entry).
3. CCTV AI video analytics.

Through a collaborative **Multi-Agent Risk Fusion Engine**, it cross-references these streams to predict and isolate "Compound Risks" (e.g., authorizing welding work near a methane valve leakage or letting workers enter storage tanks with low oxygen) before they escalate into fatal accidents. The system triggers automated mitigation loops: sounding sirens, shutting down supply lines, increasing ventilation blowers, directing workers to safety assembly points, and generating statutory compliance reports under the Factories Act 1948.

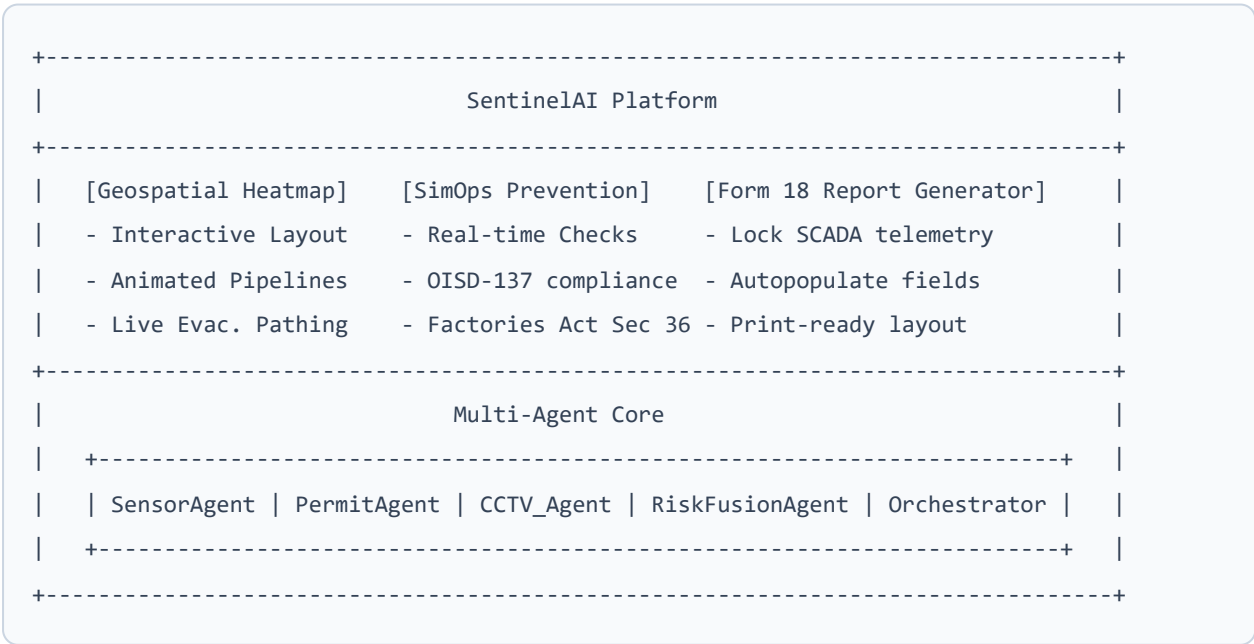
2. Problem Context & Market Gap

Heavy manufacturing sectors in India pay a devastating human cost. According to DGFASLI statistics, over 6,500 fatal workplace accidents occurred in FY2023 alone (excluding mining and construction). Investigations into past industrial accidents reveal a repeating pattern: **the problem is not the lack of safety technology, but the absence of correlation.**

- **The Silo Problem:** SCADA systems track temperature, pressure, and gas. Work Permit to Work (PTW) systems track paper forms signed by supervisors. CCTV cameras record video. If a gas valve leaks slightly in Zone A, SCADA logs a rise to 12% LEL (which is below the standalone safety threshold of 25% LEL). At the same time, a technician starts welding in Zone A under an authorized Hot Work permit. Neither system triggers an alarm on its own. The combination of these two normal events, however, is a critical ignition hazard.
- **Manual Coordination Bottlenecks:** A FICCI survey in 2024 revealed that over 60% of large factories rely on manual phone calls or handovers to sync maintenance teams and safety officers, leading to lag times that can prove fatal.
- **Warning Fatigue:** Traditional sensor alarms are binary and trigger constantly on minor fluctuations, leading to operators ignoring warning systems.

3. The SentinelAI Solution

SentinelAI overlays existing industrial infrastructure to create a unified safety brain. It provides 360-degree situational awareness through five integrated agentic capabilities:



A. Geospatial Safety Heatmap (Interactive Blueprint)

Instead of abstract grids, SentinelAI displays a clean, visual vector layout of the plant (Zone A: Coke Oven Battery, Zone B: Hydrocracker Column, Zone C: Storage Tanks).

- **Dynamic Pipe Flows:** SVG pipelines show active moving fluid lines. Under normal compliance, they flow in corporate blue. Under hazard states, they flash red and accelerate, giving operators immediate visual cues.
- **Live Worker Evacuation Tracker:** Displays live personnel avatars wandering inside their zones. Upon alarm trigger, the workers dynamically move toward the designated **Safe Assembly Point** with pulsing radar warning circles.

B. SimOps Conflict Prevention Engine (Digital Permit Agent)

The system prevents accidents from being authorized in the first place. When an operator attempts to issue a new permit:

- It checks the live sensor telemetry in that zone.
- If a conflict exists (e.g., Methane LEL > 10% for Hot Work or Oxygen < 19.5% for Confined Space), it overlays a warning banner showing specific regulatory violations (OISD-137, Factories Act Section 36), preventing the supervisor from signing off on hazardous work.

C. Multi-Agent Reasoning Shell

SentinelAI utilizes five specialized AI agents to coordinate safety workflows:

1. **SensorAgent:** Parses real-time telemetry inputs (Methane, Oxygen, Temperature).
2. **PermitAgent:** Continuously reviews active permit IDs, valid hours, and holder details.
3. **CCTV_Agent:** Monitors camera bounding boxes to verify if technicians are wearing safety harnesses and respirator lines.
4. **RiskFusionAgent:** The central brain. It evaluates logical co-occurrences of sensor warnings and permit authorizations.
5. **EmergencyOrchestrator:** Handles responses: sounds sirens, overrides ventilation blowers, shuts gas line valves, and alerts emergency rescue squads.

D. RAG Compliance Copilot & Regulation Browser

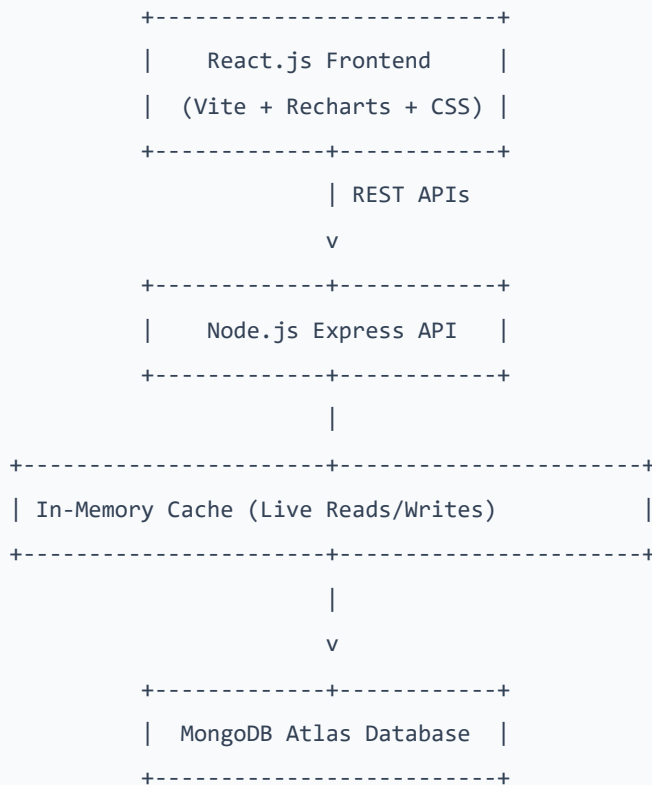
Field engineers can query the regulatory compliance database using a conversational AI interface. In addition, the interface contains a split-screen **Regulation Corpus Browser** to check section details of **OISD Standard 137** and the **Factories Act 1948** side-by-side.

E. Form 18 Incident Report Generator

On alert trigger, the platform locks all SCADA histories, active permit names, and evacuation metrics. It auto-populates a print-ready **Form 18** (the statutory accident report required under Section 88 of the Factories Act 1948) for submission to the Chief Inspector of Factories, cutting down report drafting times.

4. Technical Architecture & Tech Stack

SentinelAI is built as a highly available MERN (MongoDB, Express, React, Node.js) web application with a clean MVC controller separation:



A. Architectural Highlights

- **The Asynchronous Cache Pattern:** In a real factory, waiting for database writes over the network can cause UI lags. SentinelAI uses a local synchronous in-memory cache as its primary state. Live SCADA telemetry and logs are read and evaluated instantly from memory. Database writes are dispatched asynchronously in the background (fire-and-forget), guaranteeing sub-100ms dashboard reaction speeds.
- **MVC Separation:**
 - **Models:** Clean Mongoose schemas for `Sensor`, `Permit`, and `Alert` collections.

- **Controllers:** `dashboardController.js` and `permitController.js` handle API queries.
- **Routes:** Separate API routers under `/api`.
- **Services (`aiEngine.js`):** Contains the simulation loop and multi-agent safety evaluation functions.

5. Regulatory Compliance & Mapping

SentinelAI maps telemetry and permit criteria to statutory codes:

REGULATORY STANDARD	TARGET CONCERN	SAFETY BOUNDARY	SENTINELAI AUTOMATED ACTION
OISD-STD-137 (Part 6.2)	Hot Work Permits	Gas & Methane LEL must remain < 10%	- Flash Coke Oven node red - Block hot work permit creation - Open ventilation override - Close gas line solenoid
Factories Act 1948 (Sec 36)	Confined Space Entry	Oxygen level must remain > 19.5%	- Flash Tank storage node red - Block confined entry permit - Sound standby rescue winch alarm
Factories Act 1948 (Sec 88)	Incident Reporting	Mandatory reporting of near-misses / safety occurrences	- Lock SCADA readings on trigger - Populate Form 18 Accident Report

6. Verification & Testing Plan

To verify the correctness of SentinelAI's safety intelligence, we run testing scenarios:

Test Case 1: SimOps Violation Check

1. Select **Zone A (Coke Oven Battery)**.
2. Verify that the live Methane sensor reads above 10% LEL.
3. Open the **Permit to Work Manager** and select **Hot Work**.
4. **Expected Result:** A prominent yellow SimOps warning banner appears, citing **OISD-STD-137** safety boundaries and preventing accidental sign-off.

Test Case 2: Evacuation Action Loop

1. Verify the default state is normal (Safety Index = 100%, workers are stationary).
2. Wait for the background simulation loop to spike Methane levels in Zone A while the default **Hot Work** permit is active.
3. **Expected Result:**
 - Safety Index drops to **45% (HAZARD ACTIVE)**.
 - Zone A flashes red; the piping flows speed up and turn red.
 - **The two workers in Zone A run to the Safe Assembly Point** with pulsing red panic radar indicators.
 - A critical alarm is pushed to the Active Alarms panel.
 - scrolling agent reasoning logs output warning messages from `RiskFusionAgent` and `EmergencyOrchestrator`.

7. Business Value, Scalability, and ROI

- **Reduction of False Negatives:** By shifting from single-sensor checks to compound risk correlation, SentinelAI reduces the false negative rate, saving worker lives and preventing major plant shutdowns.
- **Lower Warning Fatigue:** Safety officers only receive alerts for validated, compound threats, increasing alarm compliance.
- **Low Integration Barrier:** SentinelAI runs as an overlay. It connects to existing SCADA database APIs, CCTV hardware streams, and digital permit databases, requiring zero downtime to install.
- **Reduced Insurance Premiums:** Demonstrating continuous compliance audits (OISD, Factories Act) helps industrial companies negotiate lower liability insurance premiums.